

MANGALORE REFINERY & PETROCHEMICALS LTD. (MRPL)

MAINTENANCE STANDARD OPERATING PROCEDURE

SOP-MNT-402 — Static Equipment Pressure Vessel Maintenance

Revision: 3 | Effective Date: 2026-01-15 | Owner: Maintenance Dept.

1. PURPOSE

This procedure defines pressure limits and mandatory response actions for pressure vessels under Maintenance jurisdiction during routine inspection, hydrotest, and in-service monitoring.

2. PRESSURE LIMITS

- Safe operating limit: up to 15 bar(g). Normal maintenance work windows and confined-space entry permits may proceed.
- Caution band: 15.1 - 17.9 bar(g). Increase monitoring frequency to every 15 minutes; notify area supervisor; suspend hot work.
- CRITICAL limit: 18 bar(g) or above. This is a SAFETY VIOLATION. Any reading at or above 18 bar(g) requires immediate isolation, evacuation of the work zone, and escalation per Section 4.

3. RESPONSE MATRIX

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|------------------|---|
| Reading < 15 bar | : Normal — proceed with planned maintenance. |
| 15 - 17.9 bar | : Caution — increased monitoring, no hot work. |
| >= 18 bar | : CRITICAL / SAFETY VIOLATION — stop work, isolate, evacuate, notify shift supervisor and Maintenance Head immediately. |

4. ESCALATION

On any CRITICAL reading (≥ 18 bar):

- Stop all work on the vessel immediately.
- Isolate the vessel and vent per the unit blowdown procedure.
- Evacuate personnel from the immediate work zone.
- Notify the Shift Supervisor and Maintenance Head within 15 minutes.
- Log the event and initiate an incident report per SOP-SAF-010.

5. RECORDS

All pressure readings during maintenance activity shall be logged in the vessel maintenance log with timestamp, gauge ID, and reading.